Data Science Foundational Level Training - NumPy

Presented By : Harsh Dalal

We Make You Trust Data

Agenda

- What NumPy is and why it is important
- Basics of NumPy, including its fundamental objects
- Create and print a NumPy array
- Carry out basic operations in NumPy
- Use shape manipulation and copying methods
- Execute linear algebraic functions
- Build basic programs using NumPy

```
distance=[10,15,17,26]
time=[.30,.47,.55,1.20]
```

Collection of values

Multiple types (heterogeneous)

Add, remove, update

Quick Recap: Lists

- A list is a collection of values. You can individually add, remove, or update these values. A single list can contain multiple data types.

```
distance=[10,15,17,26]
time=[.30,.47,.55,1.20]
```

```
speed=distance/time
```

Mathematical operation over the entire “distance” and “time” lists

```
---------------------------------------------------------------------------
TypeError                                 Traceback (most recent call last)
<ipython-input-37-b779bad68500> in <module>()
----> 1 speed=distance/time

TypeError: unsupported operand type(s) for /: 'list' and 'list'
```

Error

Limitations of Lists

- Though you can change individual values in a list, you cannot apply a mathematical operation over the entire list.

```
distance=[10,15,17,26]
time=[.30,.47,.55,1.20]
```

```
import numpy as np
```

Import NumPy

```
np_distance = np.array(distance)
np_time=np.array(time)
speed=np_distance/np_time
```

Create “distance” and “time” NumPy arrays

Mathematical function applied over the entire “distance” and “time” arrays

```
speed
```

```
array([ 33.33333333,  31.91489362,  30.90909091,  21.66666667])
```

Output

Why NumPy

- Numerical Python (NumPy) supports multidimensional arrays over which you can easily apply mathematical operations.

NumPy Overview

- NumPy is the foundational package for mathematical computing in Python.

It has the following properties:

Supports fast and efficient multidimensional arrays (ndarray)

Performs linear algebraic operations, Fourier transforms, and random number generation

Efficient way of storing and manipulating data

Executes element-wise computations and mathematical calculations

Tools for reading/writing array based datasets to disk

Tools for integrating language codes (C, C++)

Knowledge Check—Sequence it Right!

The code here is buggy. You have to correct its sequence to debug it.

1

```
distance=[10,15,17,26]
time=[.30,.47,.55,1.20]
```

2

```
np_distance = np.array(distance)
np_time=np.array(time)
```

3

```
import numpy as np
```

4

```
speed=np_distance/np_time
```

```
speed
```

```
array([ 33.33333333,  31.91489362,  30.90909091,  21.66666667])
```

Knowledge Check—Sequence it Right!

The code here is buggy. You have to correct its sequence to debug it.

1

```
distance=[10,15,17,26]
time=[.30,.47,.55,1.20]
```

2

```
import numpy as np
```

3

```
np_distance = np.array(distance)
np_time=np.array(time)
```

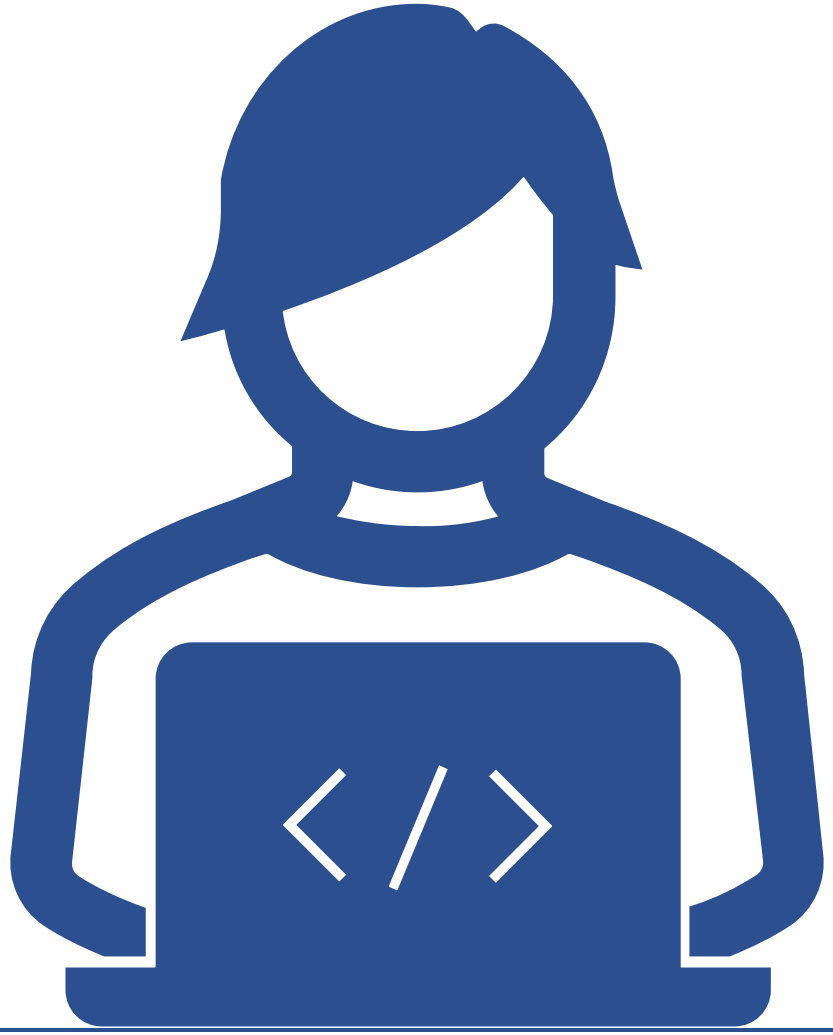
4

```
speed=np_distance/np_time
```

```
speed
```

```
array([ 33.33333333,  31.91489362,  30.90909091,  21.66666667])
```

Hands - on



Knowledge Check

- NumPy Quiz

Knowledge Check

How many elements will the following code print?

```
print(np.linspace(4,13,7))
```

a. 4

b. 7

c. 11

d. 13

Knowledge Check

How many elements will the following code print?

```
print(np.linspace(4,13,7))
```

a. 4

b. 7

c. 11

d. 13

The correct answer is : **b**

Explanation: In the “linspace” function, “4” is the starting element and “13” is the end element. The last number “7” specifies that a total of seven equally spaced elements should be created between “4” and “13,” both numbers inclusive. In this case, the “linspace” function creates the following array: [4. 5.5 7. 8.5 10. 11.5 13.]

Knowledge Check

Which of the following arrays is valid?

a. [1, 0.3, 8, 6.4]

b. [“Lucy”, 16, “Susan”, 23, “Carrie”, 37]

c. [True, False, “False”, True]

d. [3.14j, 7.3j, 5.1j, 2j]

Knowledge Check

Which of the following arrays is valid?

a. [1, 0.3, 8, 6.4]

b. [“Lucy”, 16, “Susan”, 23, “Carrie”, 37]

c. [True, False, “False”, True]

d. [3.14j, 7.3j, 5.1j, 2j]

The correct answer is : **d**

Explanation: A NumPy ndarray can hold only a single data type, which makes it homogenous. NumPy supports integers, floats, Booleans, and even complex numbers. Of all the options provided, only the array containing complex numbers is homogenous. All the other options contain more than one data type.

Summary

- NumPy is a very powerful Python library for mathematical and scientific computing. You can create and print NumPy arrays using different methods.
- Arrays can be one-dimensional, two-dimensional, three-dimensional, or multi-dimensional.
- NumPy uses basic operations, data access techniques, and copy and view techniques for datawrangling.
- NumPy can also manipulate data using various array shape manipulation techniques.
- NumPy can perform linear algebra functions to fix problematic datasets and execute mathematical operations.

Any Query

Thank You

Happy learning

Data Science Foundational Level Training - Pandas

Presented By : Harsh Dalal

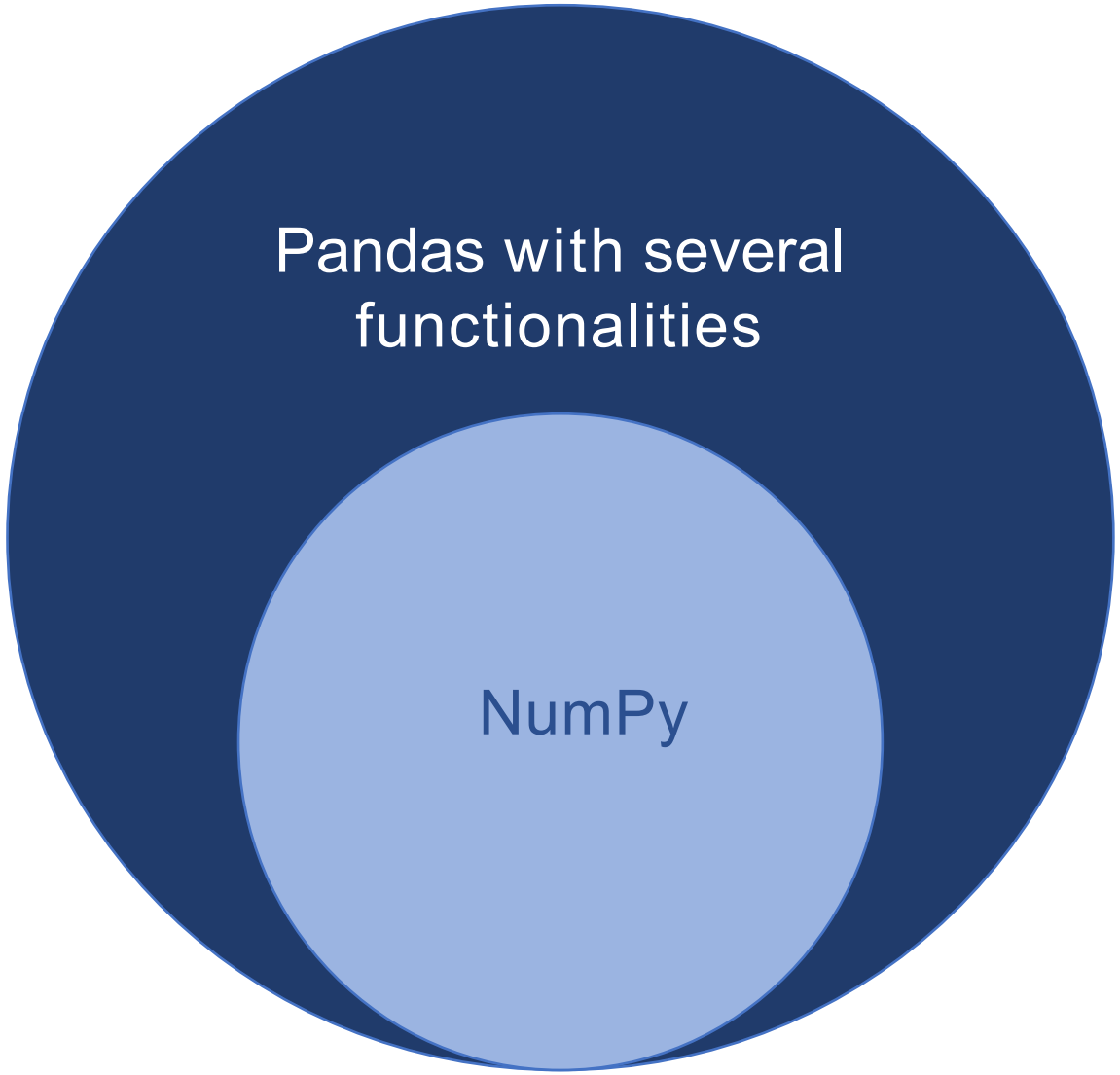
We Make You Trust Data

Agenda

- Pandas and its features
- Different data structures of Pandas
- Creating Series and Data Frame with data inputs
- Viewing, selecting, and accessing elements in a data structure
- Learning how to handle missing values
- Analyzing data with different data operation methods

Why Pandas

NumPy is great for mathematical computing. Then why do we need Pandas?



Why Pandas

NumPy is great for mathematical computing. Then why do we need Pandas?

Intrinsic data alignment

Data Structures handling major use cases

Data operation functions

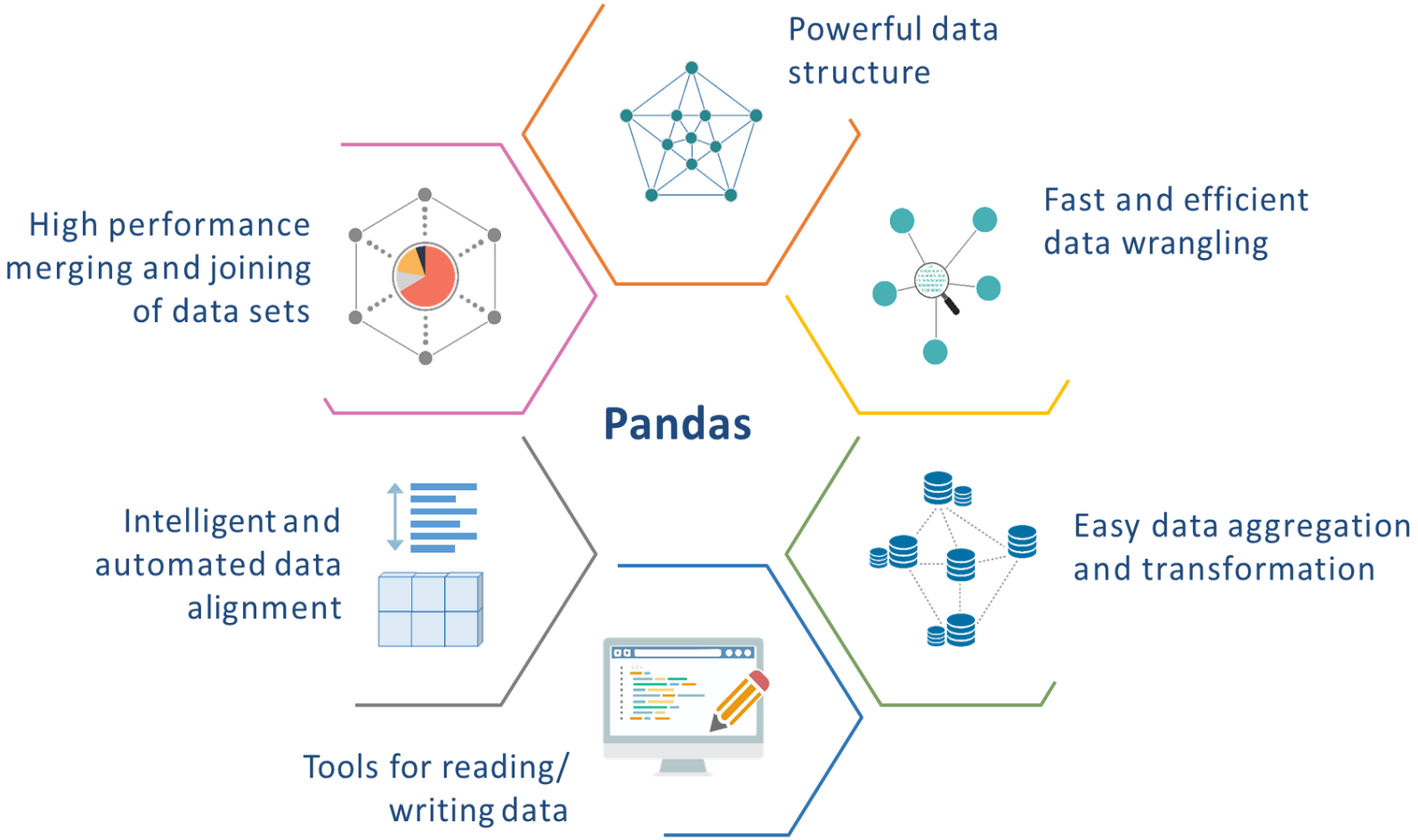
Data standardization functions

.

Functions for handling missing data

Pandas Features

The various features of Pandas makes it an efficient library for Data Scientists.



Data Structures

The four main libraries of Pandas data structure are:

Series

- **One**-dimensional labeled array
- Supports multiple data types

Panel

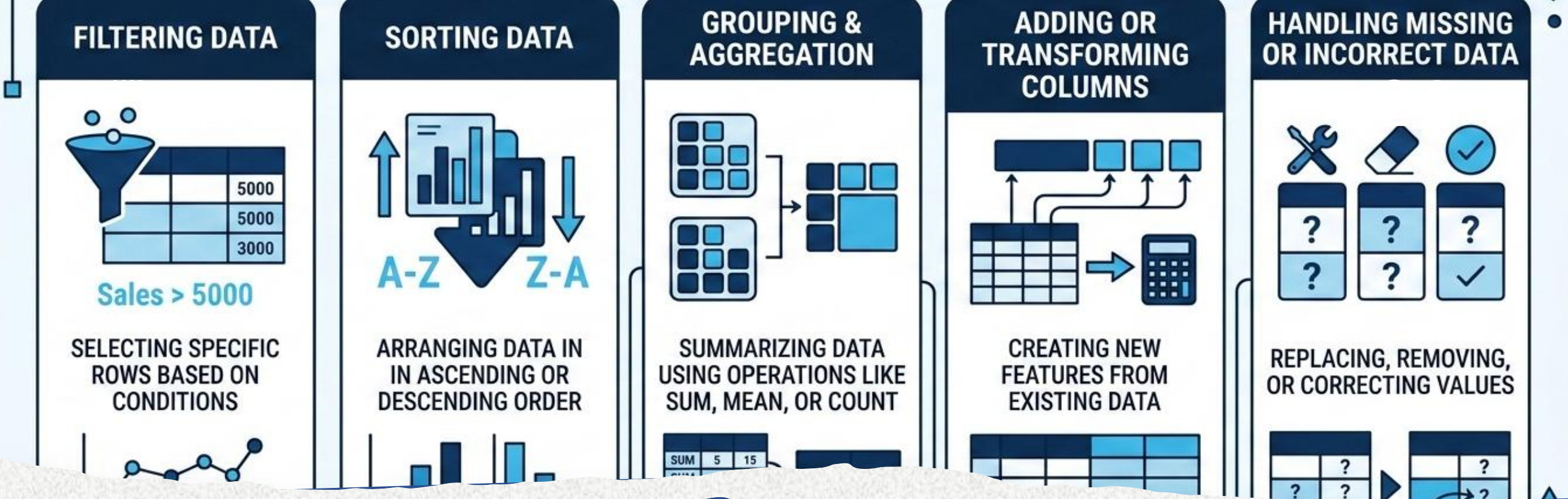
- **Three**-dimensional labeled array
- Supports multiple data types
- Items → axis 0
- Major axis → rows
- Minor axis→ columns

Data Frame

- **Two**-dimensional labeled array
- Supports multiple data types
- Input can be a Series
- Input can be another DataFrame

Panel 4D (Experimental)

- **Four**-dimensional labeled array
- Supports multiple data types
- Labels → axis 0
- Items → axis 1
- Major axis → rows
- Minor axis→ columns



Data Manipulation

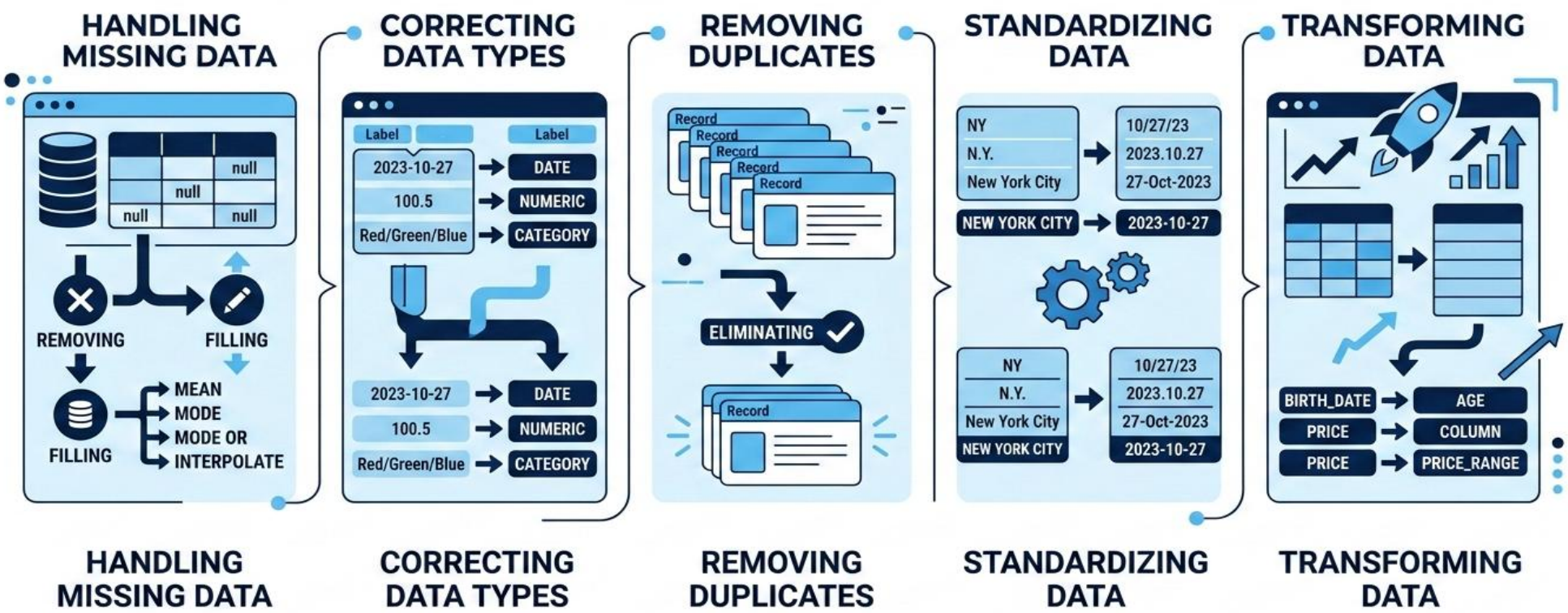
Data Manipulation refers to the process of modifying, organizing, filtering, and transforming data so it can be analyzed effectively.

Data Manipulation

```python
import pandas as pd

df = pd.read_csv("sales_data.csv")

# Filter rows
filtered = df[df["Price"] > 20000]

# Create new column
df["Total_Value"] = df["Price"] * df["Quantity"]

# Group data
summary = df.groupby("Product_Name")["Total_Value"].sum()

print(summary)
```

Data Wrangling

Data Wrangling (also called **Data Munging**) is the process of **cleaning, transforming, and structuring raw data into a usable format for analysis or machine learning.**

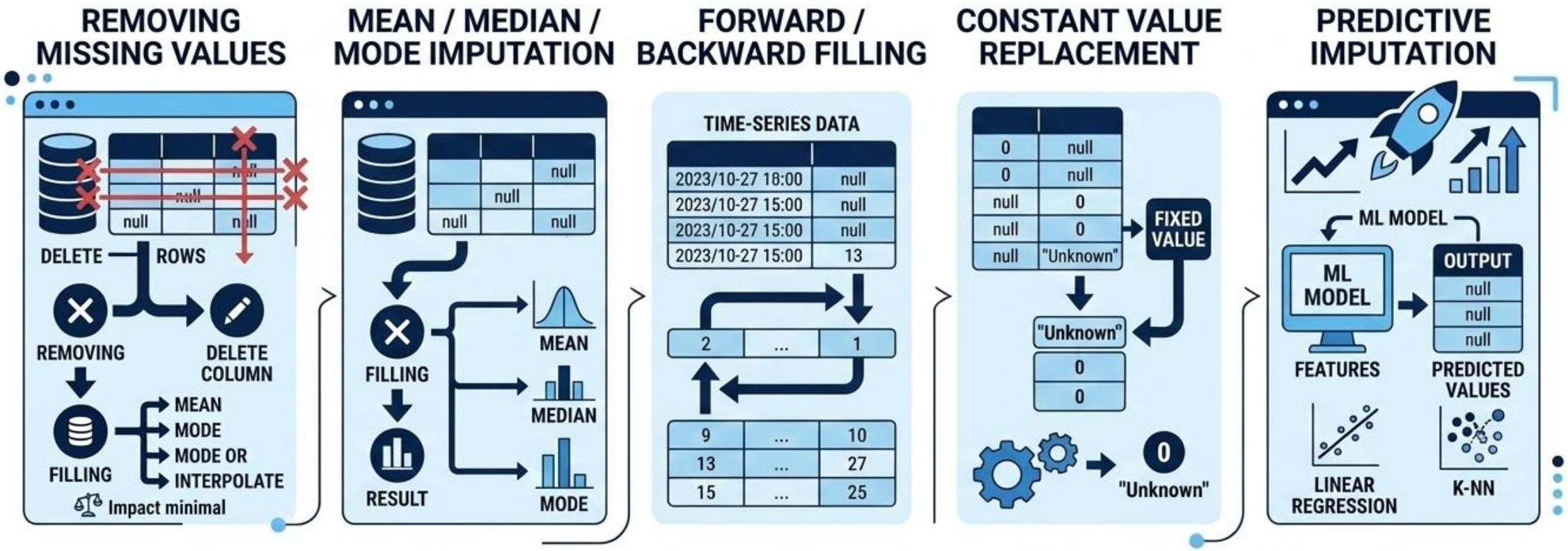


Data Wrangling

```python
import pandas as pd

df = pd.read_csv("sales_data.csv")

# Remove duplicates
df = df.drop_duplicates()

# Fill missing values
df["Price"] = df["Price"].fillna(df["Price"].mean())

# Convert data type
df["Order_Date"] = pd.to_datetime(df["Order_Date"])

print(df.head())
```

Handling Missing Data

Handling Missing Data refers to techniques used to manage **null, empty, or undefined values** in a dataset so that analysis or machine learning models remain accurate.



Handling Missing Data

```python
import pandas as pd

df = pd.read_csv("sales_data.csv")

# Check missing values
print(df.isnull().sum())

# Fill numeric column with mean
df["Price"] = df["Price"].fillna(df["Price"].mean())

# Fill categorical column
df["City"] = df["City"].fillna("Unknown")

# Drop rows with missing values
df = df.dropna()
```

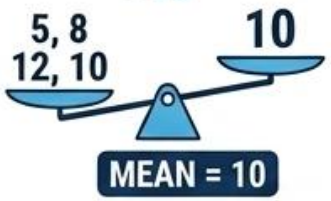
EXPLORING MEASURES OF CENTRAL TENDENCY: MEAN, MEDIAN, MODE

MEAN (AVERAGE)

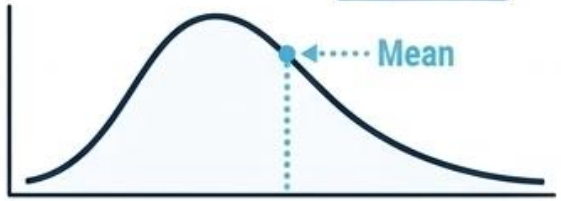
THE AVERAGE OF ALL DATA
DATA VALUES IN A SET

5
8
12
10
15

$$\frac{(5 + 8 + 12 + 10 + 15)}{5} = 10$$



MEAN = 10



MEDIAN (MIDDLE VALUE)

THE MIDDLE VALUE WHEN DATA IS ORDERED FROM SMALLEST TO LARGEST

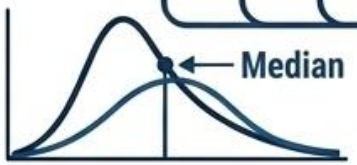
ODD NUMBER OF VALUES:
4, 7, 10, 12, 15

MEDIAN = 10

EVEN NUMBER OF VALUES: 5, 8, 11, 14, 18, 20

5, 8, 11 14 20

$$\frac{(11 + 14)}{2} = 12.5$$

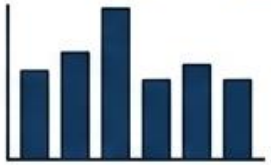


MEDIAN = 12.5

MODE (MOST FREQUENT VALUE)

THE VALUE THAT APPEARS MOST FREQUENTLY IN A DATASET

6, 9, 7, 6, 12, 6, 8, 7

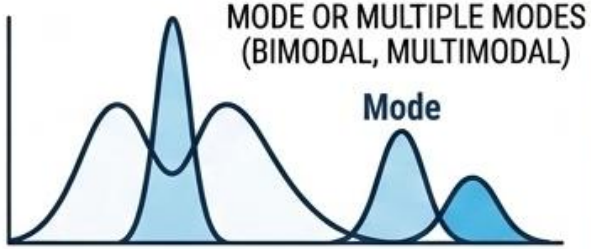


6

MODE = 6

high frequency

A DATASET CAN HAVE NO MODE OR MULTIPLE MODES (BIMODAL, MULTIMODAL)



EXPLORING MEASURES OF DISPERSION: A MEASURE OF DATA SPREAD

THE DISPERSION CONCEPT & KEY MEASURES

Range

5 8 11 15 20

Range = 15

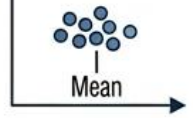
R = Max - Min

MEASURES HOW SPREAD OUT DATA POINTS ARE FROM THEIR MEAN

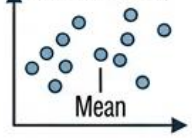
VARIANCE	STANDARD DEVIATION
$S^2 = \frac{\sum(x_i - \mu)^2}{N - 1}$	$S^3 = \sqrt{\frac{x_i - \mu)^2}{N - 1}}$

- **Variance:** Average Squared Distance from Mean
- **Standard Deviation:** Square Root of Variance (in original units)

Low Dispersion (Tight Cluster)



High Dispersion (Wide Spread)



STRUCTURE & CALCULATION OF DISPERSION

Data:

5	8	11	15	20

$$\text{Variance } (S^2) = \frac{\sum(x_i - \mu)^2}{N - 1}$$

1. Calculate Mean (μ)
2. Calculate Squared Deviations $(x_i - \mu)^2$ $(x_i - \mu)^2$
3. Sum and Divide

Example: of valuers not a small data set

$(x_i - \mu)$	$(x_i - \mu)^2$
-5 – 5	$(x_i - 1)^2$
8 – 8	$(2 - 2)^2$
11 – 15	$(3 - 3)^2$
15 – 20	- - 3

Standard Deviation (S) = $\sqrt{\text{Variance } (S^2)}$

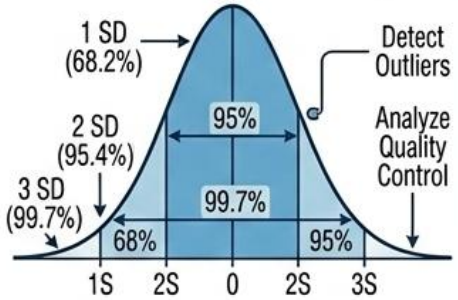
$\sqrt{S^2} = S$

Construction

1. Standardize (if needed, but less common for these measures alone)
2. Calculate deviations from mean
3. Square and sum 4. Standardize (for Sample)

INTERPRETATION & USES OF DISPERSION

Standard Deviation Range



USES:

- **Analyze Data Stability** (e.g., Finance)
- **Assess Data Representativeness** (Research)
- **Improve Quality Control** (Manufacturing)

UNDERSTANDING STANDARD DEVIATION: A MEASURE OF DISPERSION

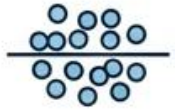
THE STANDARD DEVIATION

MEASURES HOW SPREAD OUT A DATASET IS FROM ITS MEAN

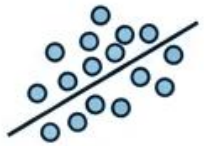
Lower standard deviation means data points are close to the mean. Higher standard deviation means data points are spread out.

5, 5.5
6, 6.5
6.5, 7

Low SD



1
3
6
9
11



High SD

CALCULATING STANDARD DEVIATION

A Multi-Step Process

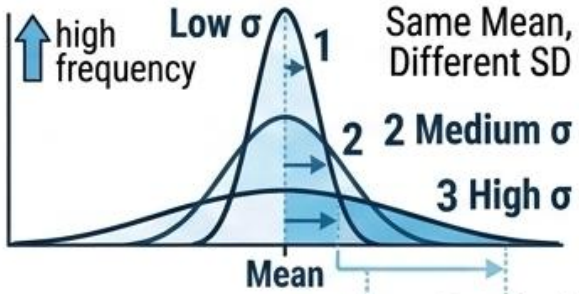
$$\sigma = \sqrt{\frac{\sum(x - \mu)^2}{N}}$$

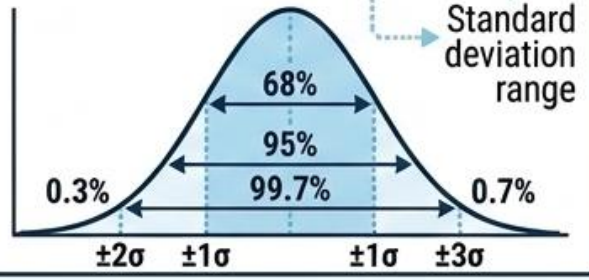
- **x** = individual value
- **μ** = mean
- **N** = number of data points
- **$(x - \mu)^2$** = squared deviations

Calculate Mean → Calculate Squared Deviations → Find Mean of Squares → Take Square Root

VISUALIZING DISPERSION

The Effect of σ on the Normal Distribution



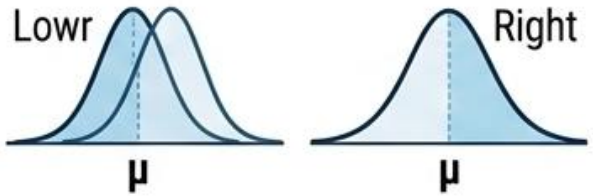


UNDERSTANDING GAUSSIAN DISTRIBUTION: THE NORMAL DISTRIBUTION

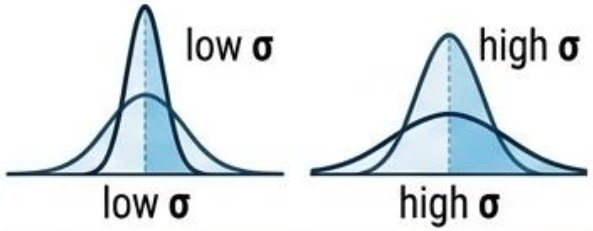
THE GAUSSIAN SHAPE

DEFINED BY **MEAN (μ)** AND **STANDARD DEVIATION (σ)**

Changes to μ (Mean)



Changes to σ (Spread)



MATHEMATICAL FORMULA & KEY TERMS

Probability Density Function (PDF) of the Normal Distribution

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

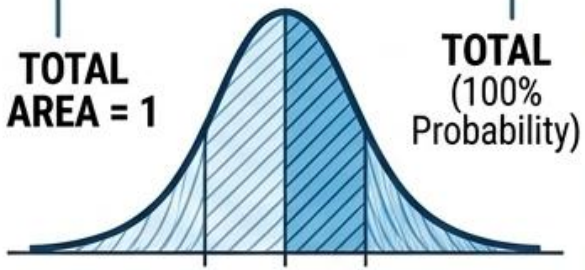
- **x** = individual value
- **μ** = mean
- **σ** = number of data points
- **π** = sound of value
- **e** = enercition

Calculate Mean → Calculate Squared Deviations → Find Mean of Squares → Take Square Root

KEY PROPERTIES & USES

SYMMETRICAL AROUND THE MEAN
(μ = Median = Mode)

TOTAL AREA = 1



TOTAL (100% Probability)

NATURAL PHENOMENA
(Heights, Errors)

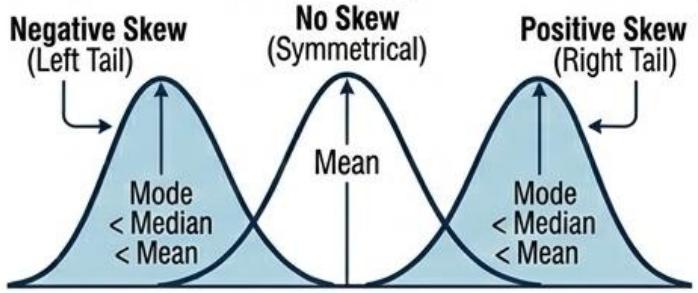
SAMPLING
(Central Limit Theorem)

MACHINE LEARNING
(Feature Scaling, Bayes Classifier)

UNDERSTANDING SKEWNESS: A MEASURE OF ASSYMETRY

THE SKEWNESS CONCEPT & INTERPRETATION

- **Symmetrical** (Mean = Median = Mode)
- **Positive Skew (Right-skewed)** (Mean > Median > Mode)
- **Negative Skew (Left-skewed)** (Mean < Median < Mode)



- Symmetrical (Mean = Median = Mode)
- Positive Skew (Right-skewed) (Mean > Median > Mode)
- Negative Skew (Left-skewed) (Mean < Median < Mode)

MEASURES HOW MUCH DATA SKEWS FROM THE CENTER

STRUCTURE & CALCULATION OF SKEWNESS

$$Skp = \frac{\text{Mean - Mode}}{\text{Standard Deviation}} = \frac{3 * (\text{Mean - Median})}{\text{Standard Deviation}}$$

If space allows, or for completeness, as Based on third moment around mean

Example calculation for Positively Skewed Data
(Data: 5, 8, 10, 15, 30)

1. Calculate Mean (μ) → 5, 8, 10
2. Calculate Standard Deviation (σ) → 30.45
3. Apply Formula → Calculated Skewness: +0.65 (Pos. Skew)

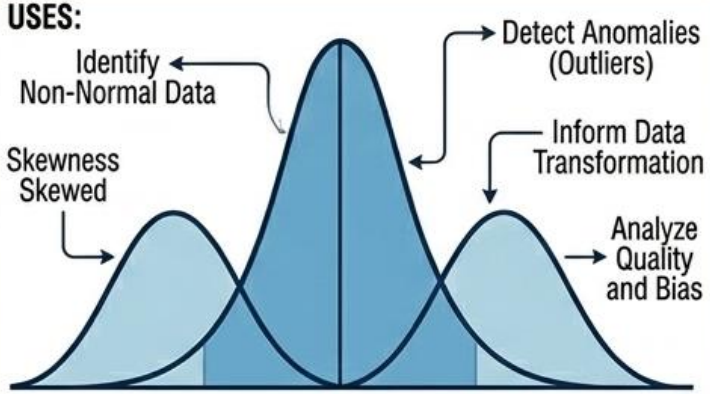
Variables - Mean
Standard Deviation

Construction

1. Standardize (if using standard formula)
2. Calculate difference between measures
3. Divide by Standard Deviation
4. Interpret result

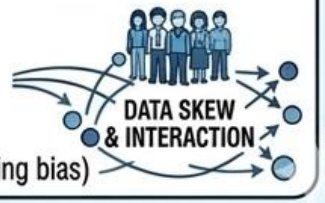
INTERPRETATION & USES OF SKEWNESS

USES:



USES:

- Inform Machine Learning Models (e.g., Log Transformation)
- Analyze Financial Returns (Detecting rare events)
- Improve Process Quality (Identifying bias)

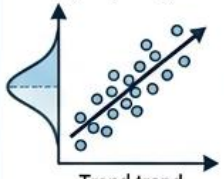


UNDERSTANDING COVARIANCE: A MEASURE OF JOINT VARIABILITY

THE COVARIANCE CONCEPT

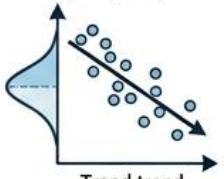
DEFINED BY **HOW TWO VARIABLES (X & Y) MOVE TOGETHER**

POSITIVE COVARIANCE (X↑, Y↑)



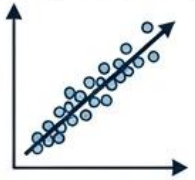
Trend trend Covariance

NEGATIVE COVARIANCE (X↑, Y↓)

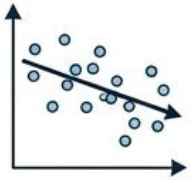


Trend trend Covariance

HIGH COVARIANCE (Strong Directional Link)



LOW COVARIANCE (Variables are Less Linked)



MATHEMATICAL FORMULA & KEY TERMS

Standard sample Covariance

$$Cov(X,Y) = \frac{\sum(X - \mu_X)(Y - \mu_Y)}{N-1}$$

- X_i = individual pair values
- μ_X = mean of X
- μ_Y = mean of Y
- N = number of data points
- $\sum($ = sum of products

$$\frac{\sum(X - \mu_X)(Y - \mu_Y)}{N - 1 \cdot \text{variables}}$$

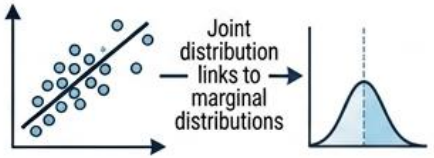
$$Cov(X,Y) = \frac{(X - \mu_X)}{n-1}\frac{(Y - \mu_Y)}{n-1}$$

$$= \frac{13 - 6)(1 - 15)}{13 - N - 1}$$

Calculate Mean of X
↓
Calculate Mean of Y
product of deviations
↓
Calculate Mean of Y
↓
Find Sum of Products
↓
Divide

KEY INTERPRETATIONS & USES

SCATTERPLOT & GAUSSIAN PROJECTIONS



COVARIANCE SIGN (POSITIVE, NEGATIVE, ZERO)

USES:

CORRELATION (Requires Standard Deviation)

CORRELATION COEFFICIENT (Requires Standard Deviation)

PORTFOLIO RISK (Asset Interaction)

MACHINE LEARNING (Feature Engineering, PCA)

EXPLORING CORRELATION MATRIX: A STRUCTURED HEATMAP OF RELATIONSHIPS

CORRELATION CONCEPT & KEY INTERPRETATION

DEFINED BY **SCALED RELATIONSHIPS (PEARSON COEFFICIENT)**

+1 (Perfect Positive)
+0.7 (Strong Positive)

-1 (Perfect Positive)	+0.7 (Strong Positive)	0 (No Linear Link)	-0.7 (Strong Negative)	+1 (Perfect Negative)

Scatterplot with upward trend	Scatterplot with downward trend	Scatterplot with no trend	Correlation Sign

MEASURES LINEAR RELATIONSHIPS BETWEEN PAIRS OF VARIABLES

Correlation
(Requires Standard Deviation)

CORRELATION COEFFICIENT
(Requires Standard Deviation)

STRUCTURE & INTERPRETATION OF A MATRIX

Correlation Matrix Example (X, Y, Z, W)

	X,X	Y,Y	Z,Z	W,W
X	+1.0	+0.85	-0.60	+0.70
Y,Y	+0.85	+1.0		+0.70
Z,Z	-0.60	+0.50	+1.0	
W,W	+0.50	-0.30		+1.0

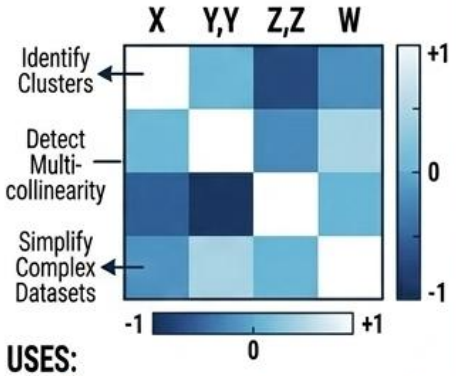
KEY FEATURES

- Diagonal is always +1
- Symmetrical (Matrix is mirrored)
- Off-diagonal values are between -1 and +1

Construction

1. Standardize variables (using Mean and SD)
2. Calculate pair-wise Pearson Correlations (ρ)
3. Arrange coefficients in a matrix form

HEATMAP VISUALIZATION & APPLICATIONS

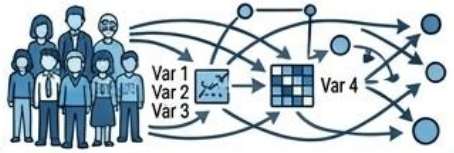


USES:

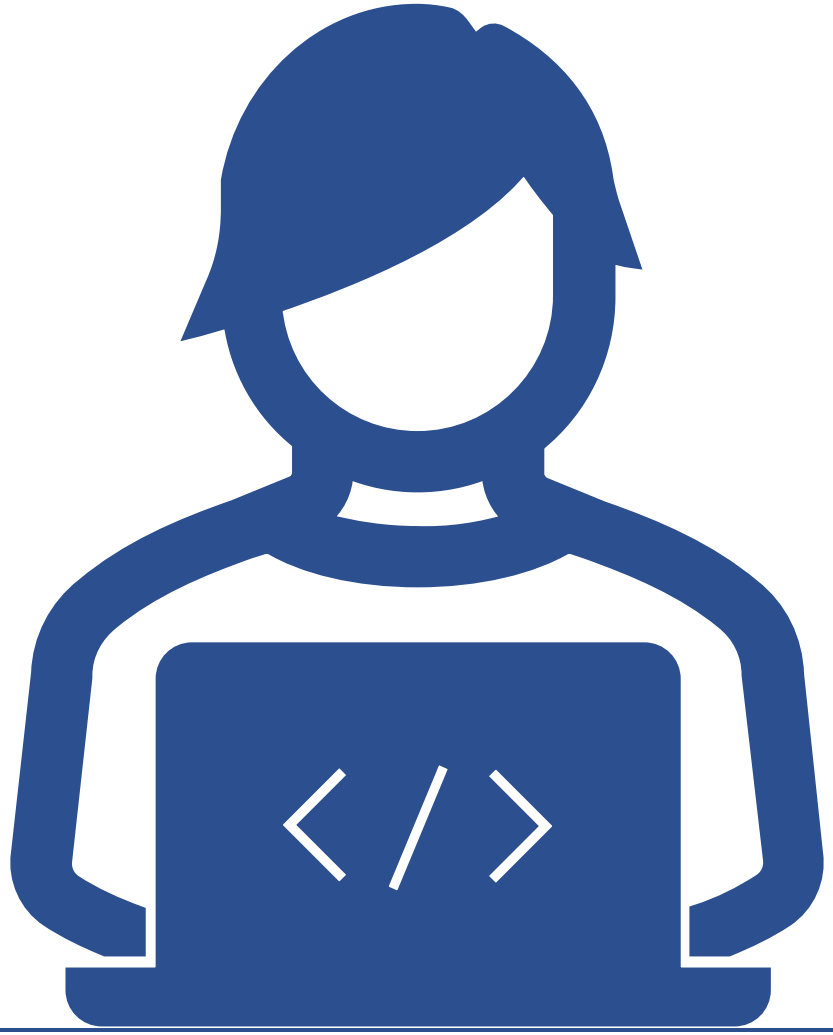
Feature Selection (ML)

Portfolio Management (Finance)

Gene Expression Analysis



Hands - on



Knowledge Check

- Pandas Quiz

Knowledge Check

How is an index for data elements assigned while creating a Pandas series ? Select all that apply.

a. Created automatically

b. Needs to be assigned

c. Once created can not be changed or altered

d. Index is not applicable as series is one-dimensional

Knowledge Check

How is an index for data elements assigned while creating a Pandas series ? Select all that apply.

a. Created automatically

b. Needs to be assigned

c. Once created can not be changed or altered

d. Index is not applicable as series is one-dimensional

The correct answer is : **a,b**

Explanation: Data alignment is intrinsic in Pandas data structure and happens automatically. One can also assign index to data elements.

Knowledge Check

What will the result be in vector addition if label is not found in a series?

a. Marked as Zeros for missing labels

b. Labels will be skipped

c. Marked as NaN for missing labels

d. Will throw an exception, index not found

Knowledge Check

What will the result be in vector addition if label is not found in a series?

a. Marked as Zeros for missing labels

b. Labels will be skipped

c. Marked as NaN for missing labels

d. Will throw an exception, index not found

The correct answer is : **C**

Explanation: The result will be marked as NaN (Not a Number) for missing labels

Summary

- Pandas is an open-source library for data analysis and is an efficient data wrangling tool in Python.
- The four main libraries of Pandas are Series, Data-Frame, Panel, and Panel 4D.
- Data-Frame is a two-dimensional labelled data structure with columns of potentially different data types.
- To access data elements in a series, 'loc' and 'iloc' methods can be used.

Any Query

Thank You

Happy learning